CURELINK

DEPI GRADUATION PROJECT

Project Documentation

Parts 1–3

Project Planning & Management • Literature Review
Requirements Gathering

TRACK

Software Development — Cross Platform Mobile App Developer

TEAM LEADER

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October 2026

github.com/Mohamedyasser9308/CureLink

Document Control

Item	Details
Project	CureLink
Document	First three required documentation parts in one submission PDF
Prepared for	DEPI Graduation Project
Team repository	https://github.com/Mohamedyasser9308/CureLink
Project planning window	2 October – 16 October 2026 (initial planning and documentation)
Submission deadline	19 October 2026 (per the team-provided submission notice)
Document status	Prepared from the team's supplied Trello export, design export, flow description, and official schedule details shared in the conversation

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Document integrity note. This report distinguishes confirmed project information from proposed targets and assumptions. No measured app outcomes, formal lecturer comments, or official grading percentages are claimed unless supplied by the team.

1. Project Planning & Management

1.1 Project Overview

CureLink is a cross-platform mobile application concept focused on medication management and caregiver support. It is intended to help patients organize medication schedules, record dose status, and review medication history, while allowing an authorized caregiver to view a linked patient's medication information and—when permission allows—help manage medicines. The interface is designed for different age groups and roles, with age-aware presentation and patient/caregiver navigation.

1.2 Problem Statement

Medication routines can be difficult to manage consistently, especially when a person has several medicines, changing schedules, or needs help from a family member or caregiver. A patient may need reminders and a simple way to record whether a dose was taken, while a caregiver may need visibility into missed doses without being given unrestricted access. CureLink addresses this problem through scheduled medication information, dose-status tracking, patient–caregiver linking, permission levels, and missed-dose notifications. The app is a medication-organization and support tool; it is not presented as a substitute for professional medical advice or clinical decision-making.

1.3 Project Objectives

- Provide a clear place for patients to add, view, edit, and delete medication records and schedules.
- Support medication reminders and dose-status recording (for example, Taken or Missed) with a history view.
- Enable patients to request caregiver access and control whether linked caregivers have read-only or medication-management permissions.
- Provide role-aware interfaces for patients and caregivers, including age-group adaptations for children, adults, and older adults.
- Support account creation, sign-in, saved profile information, session handling, and role-based navigation.
- Use accessible interface choices, localization, light/dark themes, and consistent shared components.
- Test the main flows and integrate team contributions through a shared GitHub repository and Trello planning board.

1.4 Project Scope

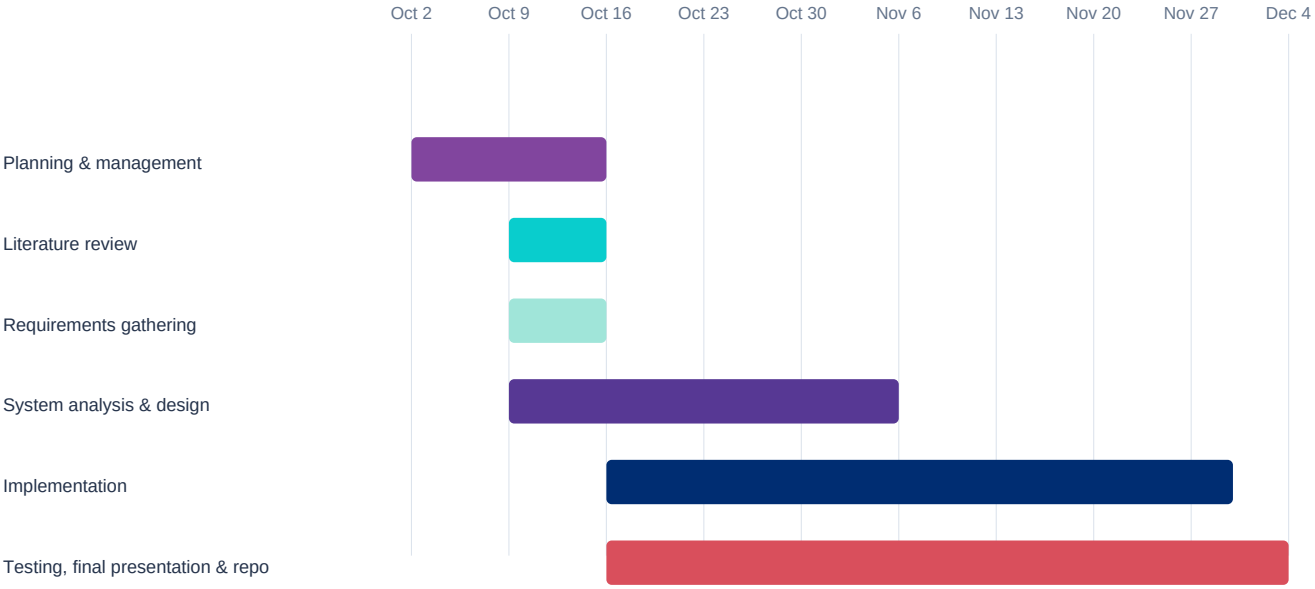
In scope for the planned application	Out of scope / not promised in this project description
Account registration and login using Firebase Authentication; saved profile and role information.	Diagnosis, prescribing, dosage recommendations, or changing a clinician's treatment plan.
Patient medication CRUD, medication details, schedules, dose status, and medication history.	Guaranteeing clinical outcomes or guaranteeing that a user will take every dose.
Patient/caregiver roles, patient linking request and approval, and explicit permission levels.	Unrestricted caregiver access to all patient data without patient approval.
Reminder and missed-dose notification flow, including caregiver alerts where linked and permitted.	Emergency response service or guaranteed SOS dispatch unless explicitly implemented and tested.
Age-group-aware UI, localization foundation, light/dark theme, loading/empty/error states.	Claiming regulatory certification, clinical validation, or production-grade reliability without evidence.
Testing, review, and merging through team workflow.	Replacing professional medical advice or emergency services.

1.5 Deliverables

Deliverable	Evidence / acceptance indication
Mobile app source code	Team repository contains the Flutter project and integrated branches.
Authentication and profile flows	Registration/login validation, account state, role/profile storage, and navigation work in testing.
Medication management flows	Add, view, edit, delete, and load medication details; validate input and handle images.
Patient and caregiver experiences	Role-appropriate screens and patient-link/permission flows are implemented according to approved scope.
Reminder and dose-status flows	Scheduled reminder and taken/missed status flow are tested; caregiver alerts are verified where included.
UI design and documentation	Figma design reference, flow diagram, first three documentation sections, final testing/report materials.

1.6 Project Plan, Timeline & Milestones

The schedule below combines the official DEPI date ranges supplied by the team with the two Trello implementation weeks visible in the board export. The high-level dates are confirmed from the supplied schedule; task-level dates beyond the visible Trello weeks are planning windows, not claims that every future task has already been scheduled.



Workstream	Official / planned period	Milestone or deliverable
Project Planning & Management	2 Oct – 16 Oct 2026	Scope, schedule, roles, risk plan, KPIs
Literature Review	9 Oct – 16 Oct 2026	Similar-solution comparison and research findings
Requirements Gathering	9 Oct – 16 Oct 2026	Stakeholders, user stories, functional and non-functional requirements
System Analysis & Design	9 Oct – 6 Nov 2026	Detailed analysis/design artifacts; flow and UI design references
Implementation (source code & execution)	16 Oct – 30 Nov 2026	Integrated app features and executable build
Final presentation, testing & reports	16 Oct – 4 Dec 2026	Test evidence, final report, presentation
First three documentation parts submission	Deadline: 19 Oct 2026	One PDF uploaded to team GitHub repository

1.7 Task Assignment & Roles

The assignments below are taken from the supplied Trello board export. Cards are organized under Week 1 and Week 2; the board assigns one named owner to each card except the Week 2 card titled “Amira galal,” whose member field is blank in the export. The card title and its checklist indicate Amira’s ownership, but this should be corrected in Trello for clean project tracking.

Team member	Week 1 assignment	Week 2 assignment	Coordination / ownership notes
Mohamed Yasser	Profile creation; profile image; age group; unique ID generation.	Empty/error states; age-group UI; dark mode; loading; review and merge code.	Team leader; organizes tasks and merges branches into the shared branch.
Ahmed Abdelbar	Project structure; theme; light/dark foundation; localization foundation; shared widgets; app colors; Firebase initialization.	Patient home UI; home medication list; next dose; home states/navigation; localization.	Foundation and patient-home work.
Hana Elganagy	Splash; three onboarding screens; Next/Skip/Get Started.	Medication model/repository; Add Medicine UI; validation; medicine image; save medicine; localization.	Onboarding and medication creation flow.
Abdelrahman Bassam	Login UI; email/password validation; Firebase Auth; login states; loading; error handling.	Edit medicine screen; validation; edit image; update medicine; testing; localization.	Authentication and edit-medication flow.
Amira Galal	Sign Up UI; Firebase registration; validation.	Medicine Details UI; load medicine; delete medicine; image handling; testing; localization.	Week 2 card title identifies Amira; member assignment is blank in export and should be corrected.
Veronica Kamil	Role selection; save role; role-based navigation.	Session management; account state; logout; session testing; navigation integration.	Role and session flows.

Team workflow

The team uses Trello to divide work into weekly cards and checklist items, and a shared GitHub repository for code integration. Each contributor works on a branch and pushes changes; the team leader reviews and merges contributions. This workflow supports parallel development, but shared interfaces, routes, models, and Firebase structures should be agreed before dependent tasks are merged.

1.8 Resource Allocation

Resource	Use in CureLink
Flutter / Dart	Cross-platform mobile application and UI implementation.
Firebase Authentication	Account registration and sign-in; authentication state.
Cloud Firestore	Planned persistence for profiles, medication records, schedules, dose logs, and caregiver links, as reflected in the supplied flow diagram.
Firebase Cloud Messaging (FCM) and Cloud Functions	Planned notification delivery and scheduled/missed-dose processing, as reflected in the supplied flow diagram; implementation and reliability must be verified in testing.
Figma design export	Shared UI reference and visual consistency across screens.
GitHub	Source control, branches, integration, review, and delivery.
Trello	Weekly task assignment, checklist progress, and team coordination.
Mermaid flow diagram	High-level account, medication, caregiver-link, and notification flow reference.

1.9 Risk Assessment & Mitigation Plan

Risk	Likelihood / impact	Mitigation / response
Firebase configuration or integration errors	Medium / High	Keep setup documented; test auth and database access early; use a shared configuration checklist; verify environment setup before merging.
Notification permission, delivery, or background execution issues	Medium / High	Test on supported devices and OS versions; handle denied permissions; document platform limits; show an in-app status where appropriate.
Role/permission logic exposes patient information incorrectly	Medium / High	Enforce authorization in data access rules, not only UI; test read-only vs manage-medications cases and patient approval states.
Merge conflicts or incompatible interfaces	High / Medium	Use feature branches, small pull requests, consistent naming, agreed models/routes, and leader review before merge.
Schedule pressure near submission deadlines	Medium / High	Prioritize critical flows; define a minimum working scope; track checklist completion; reserve time for integration and regression tests.
Localization or theme inconsistencies	Medium / Medium	Use shared theme/localization resources; avoid hard-coded strings; review light/dark and age-group screens together.
Medication data loss or incorrect status records	Medium / High	Validate inputs; handle loading/error states; verify CRUD and status transitions; avoid representing unconfirmed data as taken/missed.
Sensitive health-related information is mishandled	Medium / High	Collect only required data; apply authenticated access and least privilege; avoid logging secrets or unnecessary health details; test access rules.
Accessibility issues for children or older adults	Medium / Medium	Review contrast, readable text, touch targets, simple language, and age-aware layouts against the supplied UI design.

1.10 Key Performance Indicators (KPIs)

The following are proposed acceptance targets for project evaluation, not measured results. The team should record actual results during testing and revise targets if the course rubric or technical constraints require it.

KPI	Proposed target / measurement method
Critical-flow test pass rate	100% of defined critical-path tests pass before final demo; report passed / total tests.
Medication CRUD correctness	All planned add, view, edit, and delete test cases pass with correct persisted data.
Authorization correctness	All role and caregiver-permission test cases pass; unauthorized access attempts are denied.
Input validation	Invalid required fields and invalid schedule inputs are rejected with understandable feedback.
UI state coverage	Loading, empty, error, and success states are checked for each critical data flow.
Localization coverage	All user-facing strings in the agreed supported languages use localization resources; no critical screen has missing keys.
Theme and age-group consistency	Critical screens reviewed in light/dark modes and applicable age-group layouts; no blocking contrast or layout defects remain.
Repository integration	All team changes intended for release are merged and the final branch builds successfully in the agreed environment.
Notification flow reliability	Run a defined set of reminder/missed-dose scenarios on supported test devices and report delivery outcomes; do not claim 100% delivery without evidence.

Acceptance principle

The project should be judged on verified behavior, usability, permission safety, code integration, and documentation—not on unmeasured claims of improved adherence or guaranteed notification delivery.

2. Literature Review

2.1 Review Purpose & Approach

This review considers the medication-management problem, research on mobile medication-adherence interventions, and publicly described features of two established apps. It is a focused project review rather than a full systematic review. Sources include a World Health Organization report, peer-reviewed systematic reviews/meta-analyses, and official product pages. Product pages are used to describe advertised features, not to independently prove clinical effectiveness.

2.2 Background: Medication Adherence and Digital Support

The World Health Organization's report on adherence to long-term therapies frames adherence as a multifactorial issue that requires attention beyond simply reminding an individual. [1] Evidence reviews suggest that mobile applications may help: a 2020 systematic review and meta-analysis of nine randomized controlled trials reported higher self-reported medication adherence among app users than comparison groups, while noting the need for larger studies and further evaluation of long-term effects. [2] A later systematic review and meta-analysis covering 26 studies and 5,174 participants also reported a positive association between mobile app interventions and medication adherence, with interactive strategies, reminders, and data-sharing among the features examined. [3] These findings support exploring reminders and shared visibility, but they do not establish that CureLink itself improves adherence; that would require evaluation of the implemented app.

2.3 Similar Solutions

Solution	Publicly described strengths	Considerations / limitations for this review	Relevance to CureLink
Medisafe	Medication reminders and dose tracking; its help documentation describes caregiver-oriented options, including missed-dose alerts and dependent profiles with medication-management capabilities. [4]	Available caregiver models differ in what the caregiver can do; the team should avoid assuming that every shared-access model fits every patient. Features summarized from public help documentation, not an independent product test.	Validates the usefulness of differentiating passive visibility from delegated medication management. CureLink plans explicit read-only vs manage-medications permissions.
MyTherapy	Medication reminders, records of taken/skipped intakes, and a health diary are described on the official product page. [5]	Its broad health-tracking scope is not identical to CureLink's role- and age-group-centered design. This review does not compare clinical outcomes between apps.	Supports the value of dose history and a consolidated medication overview. CureLink additionally emphasizes role-based caregiver support and age-aware UI.
Research evidence on medication apps	Systematic reviews report potential benefits of app-based adherence support; interactive reminders and data-sharing are promising components. [2][3]	Evidence varies by study, population, intervention, and outcome measure; app usage and self-reported adherence do not prove long-term clinical benefit.	CureLink should make reminder outcomes understandable, record dose status clearly, and evaluate the flows through usability and functional tests.

2.4 Comparison and Design Implications

- **Role-aware access:** caregiver support should be consent-based and permissions should be explicit. CureLink's proposed approval flow and Read-only / Manage Medications levels are important differentiators in the project scope.
- **Actionable medication history:** recording Taken/Missed status and showing history can help patients and authorized caregivers understand recent activity. The app must distinguish a user-confirmed dose from an automatically inferred missed dose.
- **Age-aware usability:** the supplied design uses age-group-specific interfaces, including a simpler, reward-oriented child experience and larger, higher-contrast presentation for older adults. These design decisions should be checked with accessibility-focused review rather than assumed to be effective.
- **Transparent reminders:** reminder permissions, delivery limitations, and missed-dose rules should be clearly communicated. Notification delivery must be tested on supported devices and OS versions.
- **Privacy and safety:** the app should avoid implying that it prescribes treatment or replaces clinicians. Access control, minimal data collection, and safe failure states are core requirements.

2.5 Lecturer Feedback & Evaluation

Item	Documented status
Initial concept discussion	The team reported that the lecturer responded positively to the CureLink idea.
Detailed comments	No exact lecturer comments or written evaluation were supplied for this report; therefore no specific recommendation is attributed to the lecturer.
Next step	Record any later feedback accurately and update the requirements/design decisions where appropriate.

Why this wording is used: including the initial response gives the reader context, while explicitly noting that detailed comments were not recorded avoids presenting an informal impression as formal assessment.

2.6 Suggested Improvements

Priority	Improvement	Reason / evidence basis
High	Validate caregiver permissions and patient approval end-to-end.	Shared medication data is sensitive; access must match the patient's choice and role.
High	Test reminder, missed-dose, and caregiver-alert scenarios on real supported devices.	Research supports reminders as a relevant intervention component, but delivery and app behavior must be verified locally. [3]
High	Use explicit dose states and audit-friendly history.	Prevents ambiguity between a user-marked dose and a system-generated missed-dose state.
High	Add clear empty, loading, error, and retry states.	Reduces confusion when data or network operations fail.
Medium	Review contrast, text size, touch targets, and language simplicity for child and older-adult experiences.	Aligns with the supplied age-group UI concept and accessibility needs.
Medium	Run a small usability walkthrough with representative users or proxies, with consent and no real medical data.	Can uncover confusing flows before final presentation; findings should be reported as formative feedback, not clinical validation.
Medium	Document supported OS/device assumptions and notification limitations.	Sets realistic expectations and supports reproducible testing.

2.7 Final Grading / Evaluation Criteria

The team-provided guideline asks for final grading criteria but did not include an official rubric or score weights. The criteria below are therefore proposed evaluation dimensions for team readiness, not official DEPI grading percentages. The team should replace or map them to the lecturer's official rubric if one is provided.

Proposed dimension	Evidence to present
Functional completeness	Demonstrate critical user journeys from authentication through medication management and role-aware navigation.
Correctness and validation	Show validation, data persistence, dose-state behavior, and negative/error cases.
Security and privacy	Demonstrate authentication, patient consent, caregiver permission enforcement, and restricted access.
UI quality and accessibility	Show consistency with the approved design, age-aware presentation, theme behavior, and localization.
Testing quality	Present test cases, pass/fail outcomes, defects found, and fixes; distinguish tested results from planned targets.
Code quality and collaboration	Show readable structure, branch/merge history, and integrated build.
Documentation and presentation	Provide complete required documents, references, a coherent demonstration, and honest limitations.

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Needs / interests	Requirements implications
Patient — Child	Simple medication-focused interactions, clear reminders, age-appropriate feedback, and safe adult/caregiver support.	Child profile is patient-only; UI should be simple and avoid unnecessary complexity. Reward elements are motivational UI, not medical claims.
Patient — Teen/Adult	Manage medications and schedules, record dose status, view history, and choose whether/how to link a caregiver.	Patient mode is the default for a Both account; the user can switch modes as designed.
Patient — Older Adult	Readable text, strong contrast, straightforward navigation, reminders, and support when needed.	Older-adult UI should prioritize legibility, clear actions, and forgiving error handling.
Caregiver	See linked patients and relevant medication/status information; receive alerts where configured; manage medicines only when authorized.	Caregiver dashboard must show empty state when no patients are linked and enforce permission level.
Team / developers	Clear task ownership, agreed models/routes, testable acceptance criteria, and predictable integration.	Trello ownership, Git branches, review/merge workflow, and documented interfaces are needed.
Lecturer / evaluator	Understand the problem, scope, rationale, design, implementation progress, testing, and limitations.	Provide traceable requirements, research references, demo scenarios, and evidence rather than unverified claims.
Firebase / notification services (technical dependencies)	Authenticated access, data persistence, notification delivery, and controlled backend operations.	Configuration, security rules, token lifecycle, network failures, and platform permissions must be tested.

3.2 User Types, Roles, and Access Rules

Account type / mode	Capabilities and constraints captured in the supplied project context
Child	Patient-only experience. Medication-focused screens, simple interactions, and optional stars/badges/progress as UI feedback.
Teen/Adult — Patient	Manage own medication records, schedules, dose status, history, and caregiver access requests.
Teen/Adult — Caregiver	Use caregiver dashboard to see approved linked patients and information permitted by each link.
Teen/Adult — Both	Supports patient and caregiver modes; defaults to Patient Mode and offers a mode switch.
Older Adult	Patient-only experience with larger text and higher contrast.
Caregiver account	Caregiver experience; no linked patients should produce a clear empty state. Access is limited to approved patient links and assigned permission level.

Caregiver link states: Access Request → Patient Approval → Connected. Permission options: Read-only or Manage Medications. The interface must not imply access before approval.

3.3 User Stories and Acceptance Criteria

US-01 — Account registration

As a new user, I want to create an account so that I can access the app.

Acceptance criteria: Given valid required details, when I submit registration, then an account is created and the app proceeds to role/profile setup. Invalid inputs show actionable validation messages.

US-02 — Login

As a registered user, I want to sign in so that I can reach my saved information.

Acceptance criteria: Given valid credentials, when I sign in, then my saved account/profile is loaded and I am routed to the correct experience. Invalid credentials show an error without exposing sensitive details.

US-03 — Role and profile setup

As a user, I want to select the role and age group applicable to me so that the app shows the correct navigation.

Acceptance criteria: The selected role and applicable age group are saved; Child and Older Adult are patient-only; Both is available only for Teen/Adult and defaults to Patient Mode.

US-04 — Add medication

As a patient or authorized caregiver, I want to add a medication so that its details and schedule can be tracked.

Acceptance criteria: Required fields are validated; a valid medication is saved; success or failure is communicated; the record appears in the relevant list.

US-05 — View medication details

As an authorized user, I want to open a medication record so that I can review its details and schedule.

Acceptance criteria: The selected record is loaded correctly; loading, missing-record, and error states are handled; the displayed image does not break the screen if unavailable.

US-06 — Edit medication

As a patient or caregiver with management permission, I want to update a medication so that the record remains current.

Acceptance criteria: Changes are validated and persisted; read-only users cannot edit; the updated details appear after save.

US-07 — Delete medication

As a patient or caregiver with management permission, I want to delete a medication so that obsolete records are removed from active management.

Acceptance criteria: The user receives a clear confirmation opportunity; deletion is applied only to the intended record; failures are communicated and the list remains consistent.

US-08 — Record dose status

As a patient, I want to mark a scheduled dose as taken or missed so that my medication history reflects my actions.

Acceptance criteria: The selected dose and date/time are recorded correctly; repeated taps or retries do not silently create conflicting status records.

US-09 — Receive reminder

As a patient, I want reminders for scheduled doses so that I can be prompted at the appropriate time.

Acceptance criteria: When notifications are permitted and supported, the reminder uses the configured schedule; denied permission or delivery limitations are handled transparently.

US-10 — Link a caregiver

As a patient, I want to request caregiver access and approve it so that I can share selected medication information safely.

Acceptance criteria: A request is not treated as connected until approved; the chosen permission level is stored and applied to access.

US-11 — Caregiver overview

As an approved caregiver, I want to see my linked patients and permitted medication information so that I can support them.

Acceptance criteria: Only approved links are displayed; read-only access blocks medication changes; Manage Medications allows only the authorized operations.

US-12 — Missed-dose alert

As an authorized caregiver, I want an alert for a linked patient’s missed dose when configured so that I can follow up.

Acceptance criteria: The alert is associated with the correct patient/dose and respects the link and notification settings; failures are logged/handled without claiming delivery when unconfirmed.

US-13 — Logout and session state

As a user, I want to log out so that my account is not left open to the next person using the device.

Acceptance criteria: Logout clears the active authenticated session and returns to the expected entry screen; reopening the app follows the current session state.

US-14 — Accessible presentation

As a user, I want the interface to fit my age group and display preference so that key actions are understandable.

Acceptance criteria: Applicable screens respect the selected age-group design and light/dark theme; critical text remains readable and key controls are accessible.

3.4 Use Cases

Use case	Primary actor	Preconditions	Main outcome
Register and configure profile	New user	App is installed; user is not signed in.	Account, role, and required profile details are saved; user is routed to the correct home experience.
Sign in / restore session	Existing user	Account exists.	Authentication state is checked; saved profile and role determine navigation.
Manage medication	Patient or authorized caregiver	User is signed in; caregiver permission allows management if applicable.	Medication is created, viewed, updated, or deleted; validation and errors are handled.
Record dose status	Patient	Medication schedule contains a relevant dose.	Dose status and history are updated consistently.
Request/approve caregiver access	Patient and caregiver	Both accounts exist.	Request is approved or rejected; only approved links grant the selected permission.
Review patient information	Approved caregiver	A patient link exists.	Caregiver sees permitted patient, medication, and status information.
Handle missed-dose workflow	System and, where linked, caregiver	Schedule and reminder logic are configured.	The system follows the defined grace-period/status rules and notifies an authorized caregiver when configured.
Logout	Signed-in user	Valid session exists.	Session ends and protected screens are no longer accessible.

3.5 Functional Requirements

ID	Requirement
FR-01	The system shall allow a new user to register and an existing user to sign in using Firebase Authentication.
FR-02	The system shall validate required registration/login inputs and show loading, success, and error feedback.
FR-03	The system shall save and load the user profile and selected role; it shall route the user according to the saved role and applicable age group.
FR-04	The system shall support the defined account types and role constraints: Child and Older Adult are patient-only; Teen/Adult can be Patient, Caregiver, or Both.
FR-05	For a Both account, the system shall default to Patient Mode and provide a way to switch between patient and caregiver modes.
FR-06	The system shall allow authorized users to create medication records with validated details and optional image handling.
FR-07	The system shall display medication details and handle loading, missing-record, and retrieval-error states.
FR-08	The system shall allow authorized users to update medication details and persist valid changes.
FR-09	The system shall allow authorized users to delete the intended medication record and keep the active list consistent.
FR-10	The system shall support medication schedules and show relevant upcoming/next-dose information.
FR-11	The system shall allow a patient to record a dose as Taken or Missed and maintain a history of dose status.
FR-12	The system shall request/handle notification permission and support dose reminders within platform capabilities.
FR-13	The system shall support caregiver access requests, patient approval/rejection, and connection state.
FR-14	The system shall enforce Read-only and Manage Medications permissions for each approved caregiver–patient link.
FR-15	The caregiver experience shall show linked patients and a clear empty state when none are connected.
FR-16	Where implemented and configured, the system shall process missed-dose events and send notifications to authorized linked caregivers.
FR-17	The system shall support logout and correctly handle current authentication/session state on app launch.
FR-18	The system shall provide appropriate loading, empty, error, and success states for critical data operations.
FR-19	The system shall support the agreed localization resources and light/dark theme.
FR-20	The system shall provide age-group-aware presentation consistent with the approved design.

3.6 Non-Functional Requirements

ID	Quality attribute	Requirement / verification approach
NFR-01	Security & privacy	Protected data must require authentication and appropriate authorization. Test database rules and caregiver access at the data layer, not only by hiding UI controls.
NFR-02	Reliability	Critical medication operations should show clear success/failure outcomes and recover gracefully from network or service errors.
NFR-03	Usability	Primary actions should be understandable with clear labels, feedback, and predictable navigation.
NFR-04	Accessibility	Text contrast, text size, touch targets, and layouts should be reviewed for the applicable age groups and supported display modes.
NFR-05	Performance	Common screens and medication lists should load without avoidable blocking; loading states should be shown for operations that take noticeable time. Actual response-time targets should be set after baseline testing.
NFR-06	Maintainability	Shared widgets, theme values, localized strings, and clear feature boundaries should reduce duplicated UI logic and inconsistent styling.
NFR-07	Compatibility	The app should be tested on the agreed Android/emulator and target devices; the supported OS/device range must be documented by the team.
NFR-08	Data integrity	Medication and dose-status changes must apply to the correct record and avoid inconsistent duplicate or stale UI state.
NFR-09	Localization	User-facing text should use localization resources; supported languages and fallback behavior should be specified by the team.
NFR-10	Notification transparency	The app must not imply a reminder was delivered or seen unless that status is actually confirmed; permission denial and platform restrictions should be handled.
NFR-11	Testability	Critical flows should have documented positive, negative, permission, and error-path test cases with reproducible results.
NFR-12	Safety boundaries	The app shall present medication organization/support features and shall not generate diagnosis or prescribing recommendations within the documented scope.

3.7 Data and Permission Rules

- Every medication record must be associated with the appropriate patient/profile so that a user cannot accidentally view or modify another person’s medication data.
- Caregiver access must follow the state sequence Access Request → Approval → Connected. A pending or rejected request must not grant access.
- Read-only caregivers may view only the permitted information; they must not add, edit, or delete medication records.
- Manage Medications caregivers may perform medication operations only for patients who approved that permission.
- Dose status must be represented consistently: a patient-confirmed Taken status should be distinguishable from a system-generated Missed status.
- The implementation should use secure data-access rules and test unauthorized access independently of the visual interface.

3.8 Assumptions, Constraints & Open Decisions

Type	Statement
Confirmed project context	CureLink is a cross-platform mobile app for medication management and caregiver support. The team uses Flutter, Firebase-related services, Figma, GitHub, Trello, and a Mermaid flow diagram.
Flow-derived technical plan	The supplied flow references Firebase Authentication, Cloud Firestore, Firebase Cloud Messaging (FCM), and Cloud Functions. The final implementation and service configuration should be verified in code.
Constraint	The official grading weights/rubric were not included in the materials reviewed; no percentage allocation is invented in this document.
Constraint	Detailed lecturer feedback was not recorded in the supplied context; only a positive initial reaction to the idea was reported.
Open decision	Confirm exact supported languages, minimum OS versions, device support, and whether all notification/cloud-function flows are implemented for the final build.
Open decision	Confirm final grace-period logic and exact criteria by which the system marks a dose Missed.
Open decision	Confirm whether the emergency SOS concept in the UI design is included in the functional implementation scope; it is not treated as a guaranteed emergency service in this report.
Open decision	Correct the missing Week 2 Trello member assignment on the “Amira galal” card for accurate ongoing project tracking.

3.9 Requirements Traceability (summary)

Project objective	Related requirements	Representative test evidence
Secure accounts and role-aware navigation	FR-01–FR-05, FR-17; NFR-01	Register/login tests, saved-role routing, logout, unauthorized-access checks.
Medication management	FR-06–FR-10; NFR-02, NFR-08	Add/view/edit/delete tests, validation tests, persistence and error-path checks.
Dose reminders and history	FR-10–FR-12, FR-16; NFR-10	Schedule and dose-state scenarios, permission-denied tests, notification outcome logs.
Caregiver collaboration	FR-13–FR-16; NFR-01	Pending/approved/rejected link tests; read-only vs management permission tests.
Accessible, consistent interface	FR-18–FR-20; NFR-03–NFR-06, NFR-09	Review light/dark screens, age-group UI, localization coverage, loading/empty/error states.

References

[1] World Health Organization. (2003). *Adherence to Long-Term Therapies: Evidence for Action*. WHO. https://iris.who.int/bitstream/handle/10665/42682/9241545992.pdf

[2] Armitage, L. C., Kassavou, A., & Sutton, S. (2020). Do mobile device apps designed to support medication adherence demonstrate efficacy? A systematic review of randomised controlled trials, with meta-analysis. *BMJ Open, 10*(1), e032045. https://bmjopen.bmj.com/content/10/1/e032045

[3] Effectiveness of Mobile Health Intervention in Medication Adherence: A Systematic Review and Meta-Analysis. (2025). *PubMed*. https://pubmed.ncbi.nlm.nih.gov/39821698/

[4] Medisafe Help Center. (n.d.). *Caregiver Guide: How to Use Medisafe to Manage a Family Member's Medications*. https://help-center.medisafe.com/en/articles/11793638-caregiver-guide-how-to-use-medisafe-to-manage-a-family-member-s-medications

[5] MyTherapy. (n.d.). *Medication Reminder and Pill Tracker App*. https://www.mytherapyapp.com/

[6] U.S. Food and Drug Administration. (2022; page updated subsequently). *Policy for Device Software Functions and Mobile Medical Applications*. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/policy-device-software-functions-and-mobile-medical-applications

[7] CureLink team. (2026). *CureLink mobile UI library* [Figma design export supplied by the team].

[8] CureLink team. (2026). *CureLink Trello board export* [Project task assignments and checklists supplied by the team]. https://trello.com/b/LSPhKFvL/curelink

[9] CureLink team. (2026). *CureLink source repository*. GitHub. https://github.com/Mohamedyasser9308/CureLink

Source-use note

The literature review summarizes the sources rather than reproducing them. Product feature descriptions are attributed to the products' own pages; research findings are attributed to the relevant reviews. The proposed requirements and KPIs are project-specific synthesis and should be validated by the team against the implemented code and any official course rubric.

Appendix A. Figma Design Reference

Complete UI library supplied by the team

The complete CureLink mobile UI library provided in the team design export is reproduced in the following pages. The original Figma canvas is a single very tall page, so it is split vertically across seven A4 pages for readability. No design content is intentionally omitted or replaced; the split is only a presentation format.

Source files supplied with the design export

- CureLink mobile UI library.pdf
- ONB-01-Light.pdf
- ONB-02-Light.pdf
- ONB-03-Light.pdf

CURELINK · MOBILE PRODUCT REFERENCE

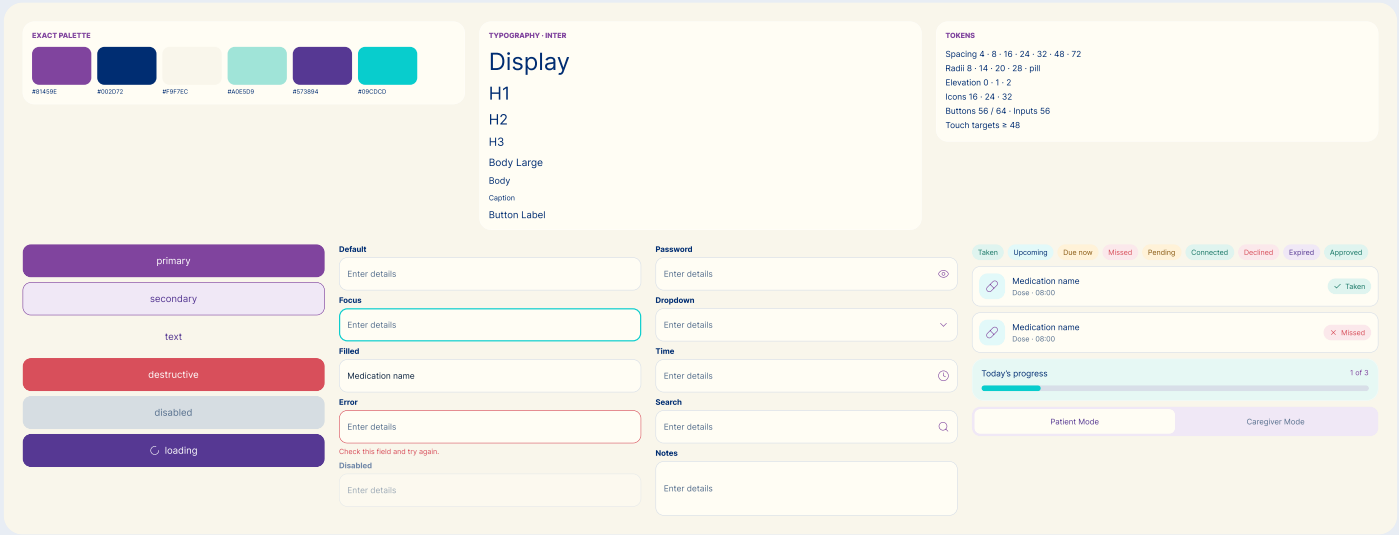
Medication care, made calm and connected.

A complete light and dark mobile screen library for patients, families, and caregivers. Structured for direct translation into reusable Flutter layouts.



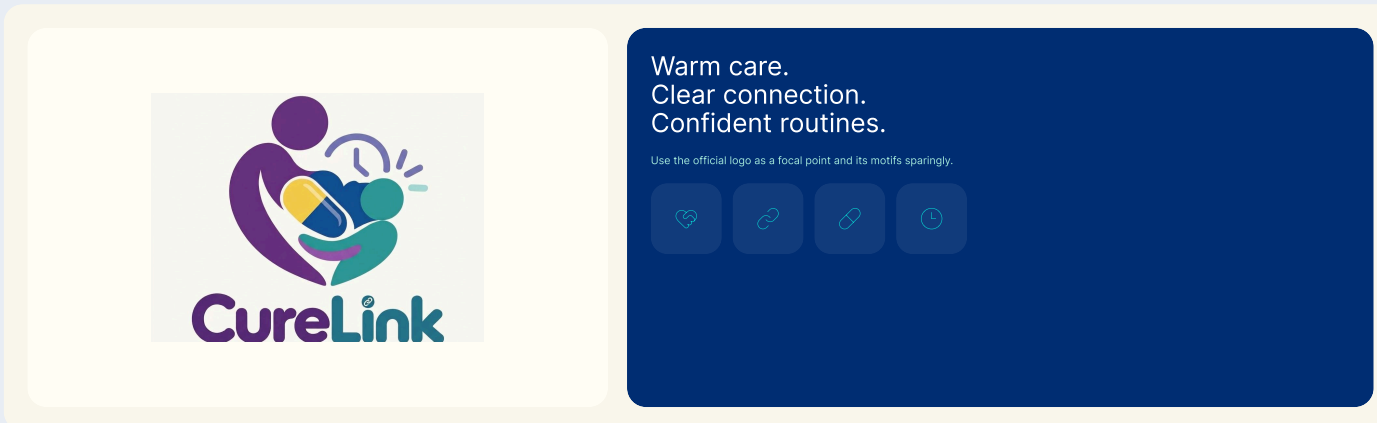
01 Design System

Foundations, accessible states, and reusable Flutter-ready patterns.



02 Brand / Logo

Official identity and rounded care motifs.

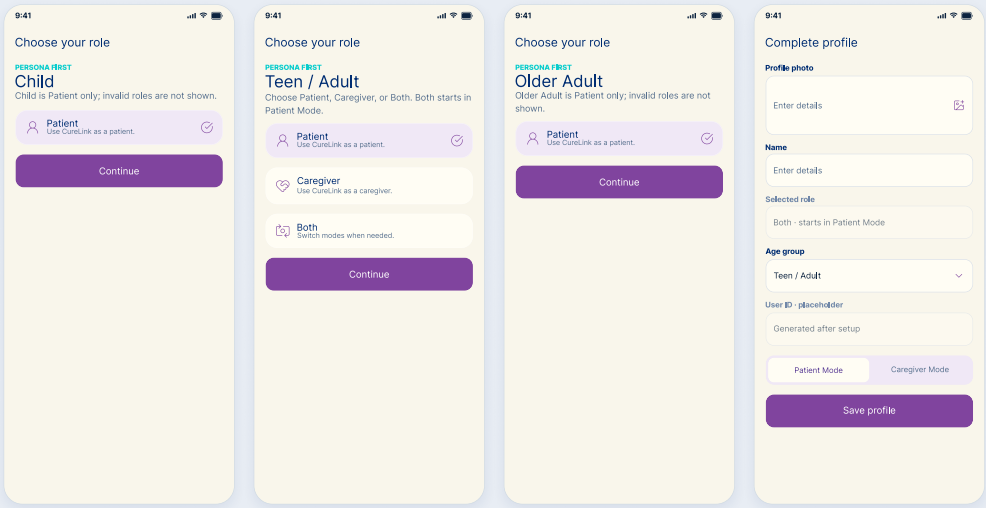


03 Splash

Official logo focal point.

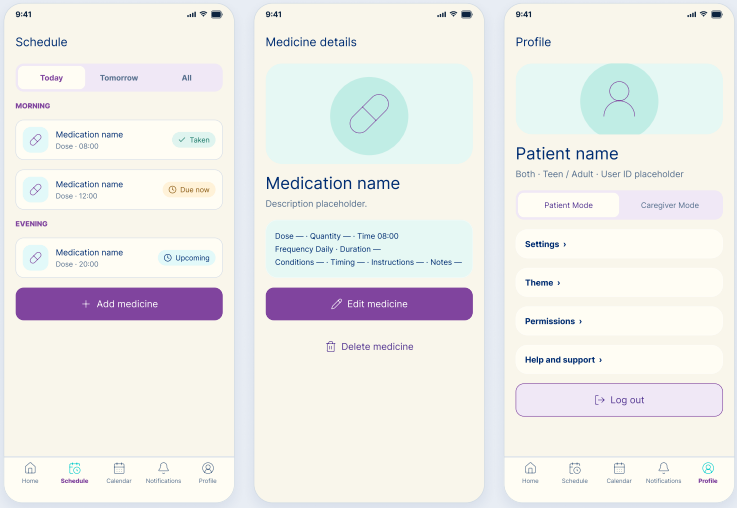
06 Persona & Role Setup

Only valid age-group roles appear.



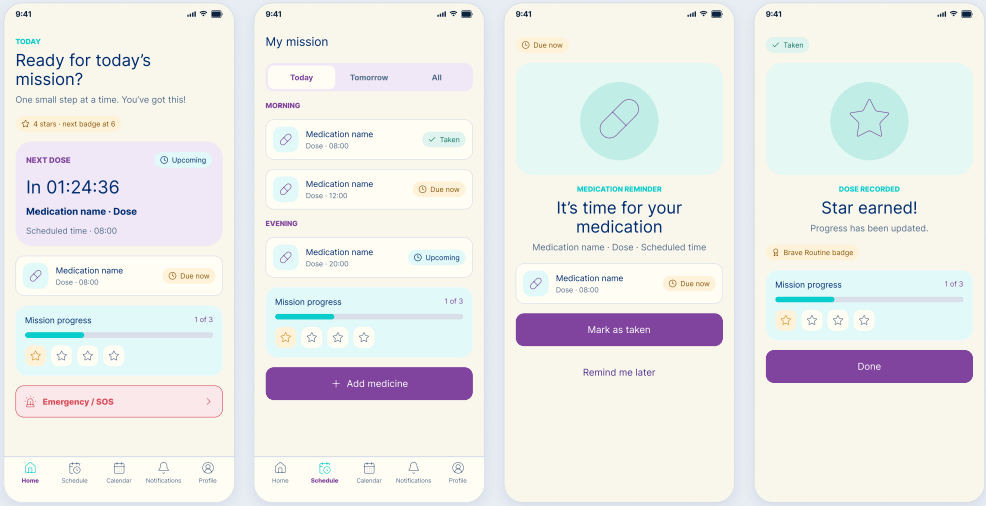
07 Patient Shared Screens

Shared navigation and reusable care structures.



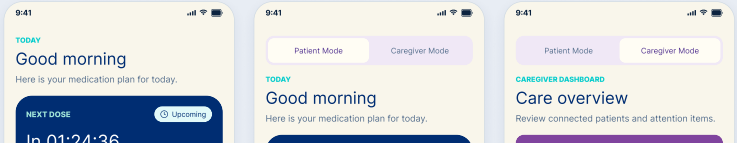
08 Child Patient

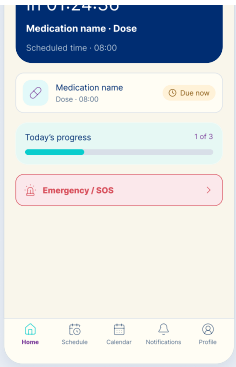
Lively missions, stars, helpers, and visible timing.

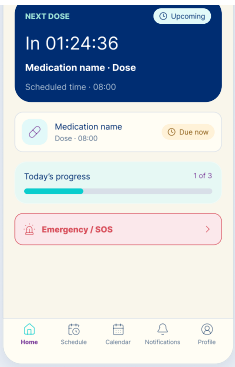


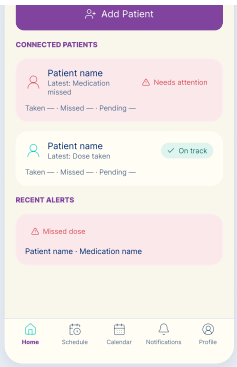
09 Teen / Adult Patient

Modern, balanced, and efficient.



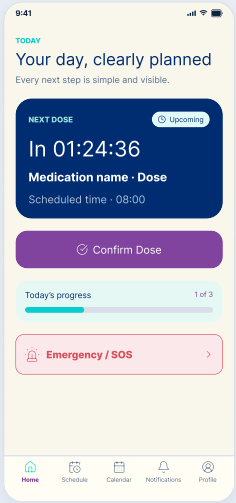


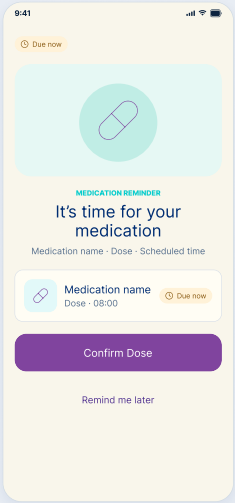


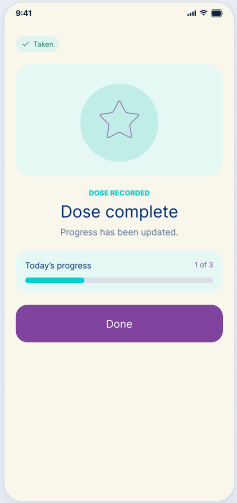


10 Older Adult Patient

Largest type, reduced density, and separated actions.

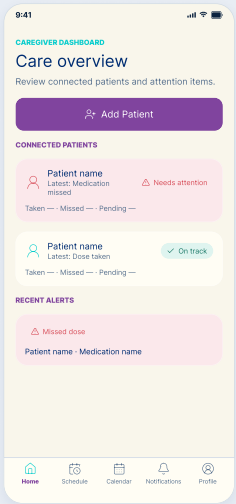


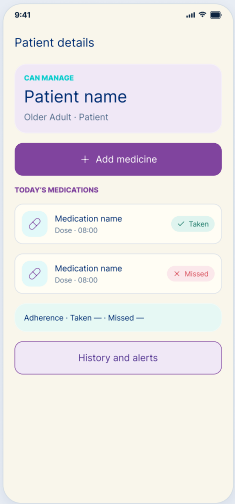


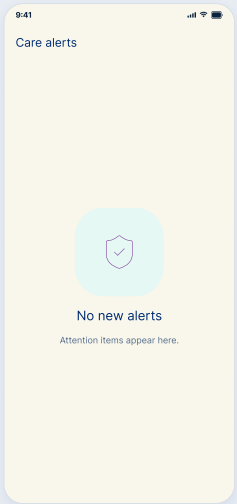


11 Caregiver

Professional monitoring, alerts, and patient identity.

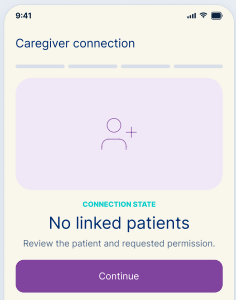


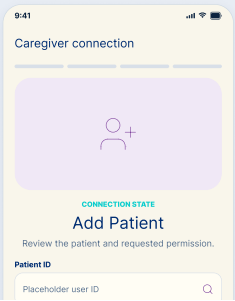


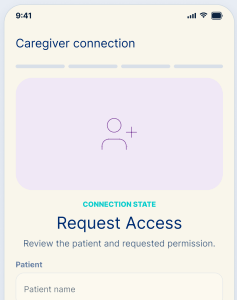


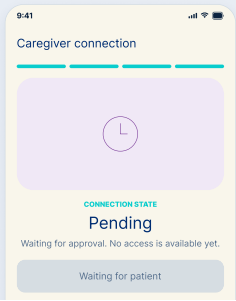
12 Caregiver Connection States

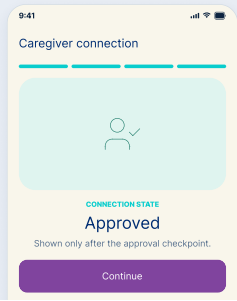
Connected appears only after approval.

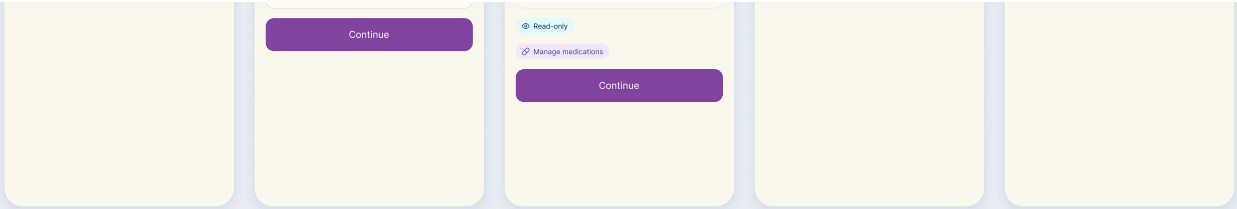


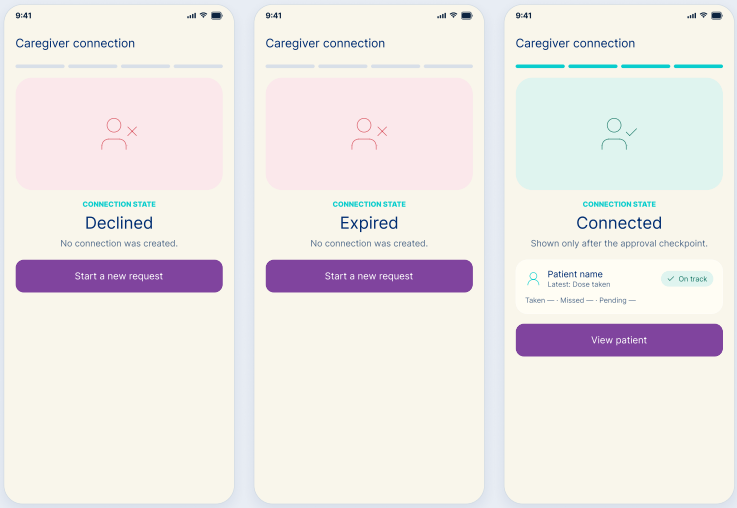






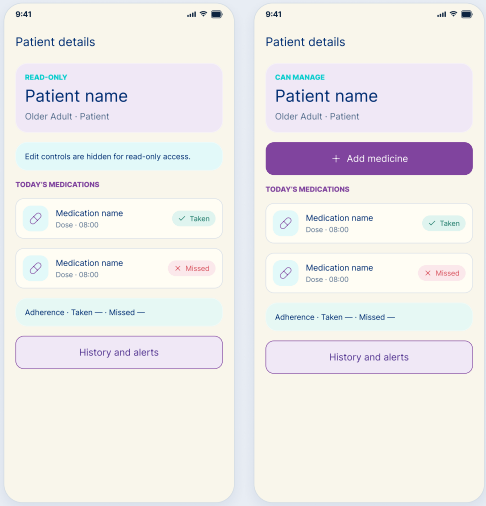






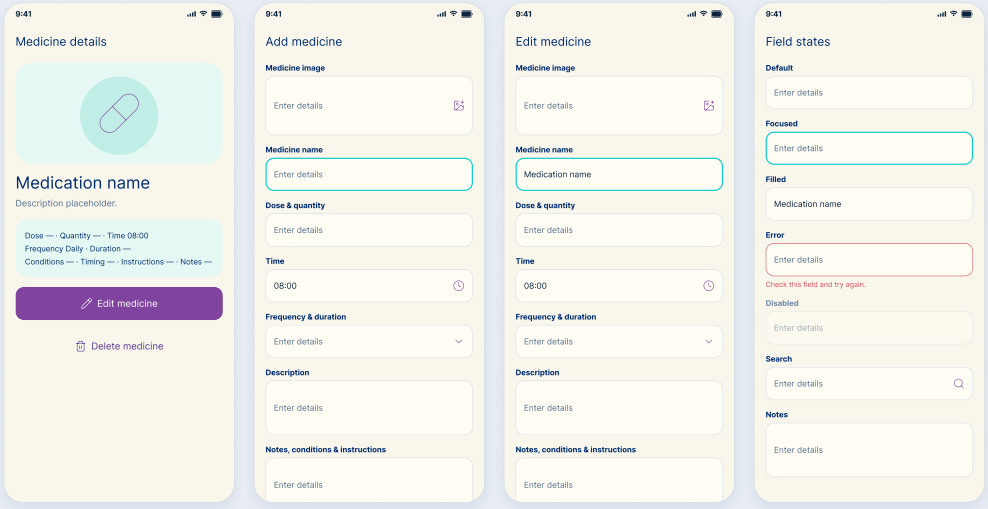
13 Permission States

Read-only and medication management.



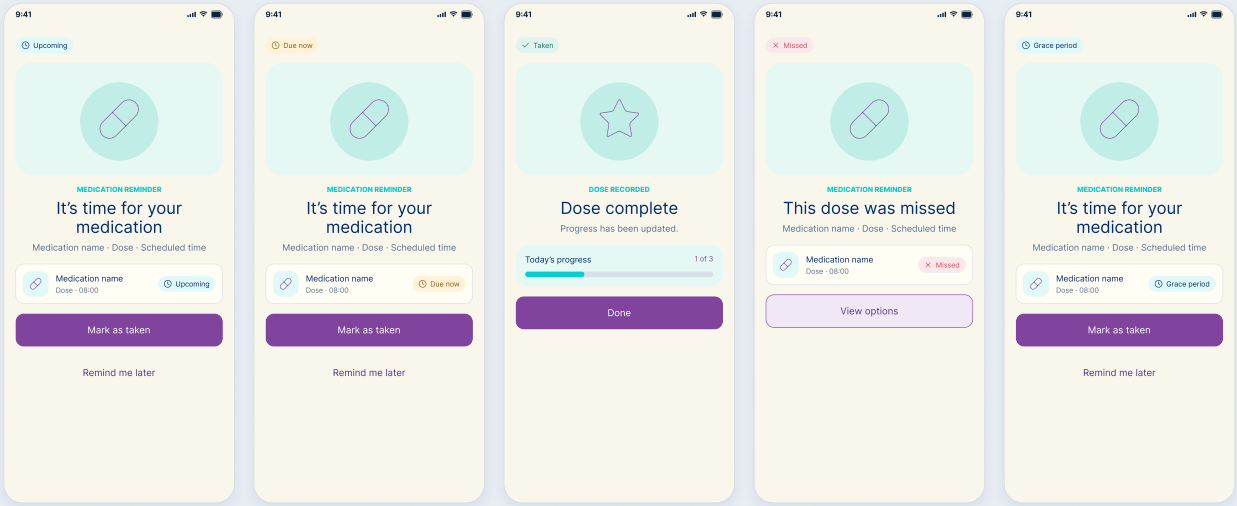
14 Medication Management

Details and complete reusable forms.



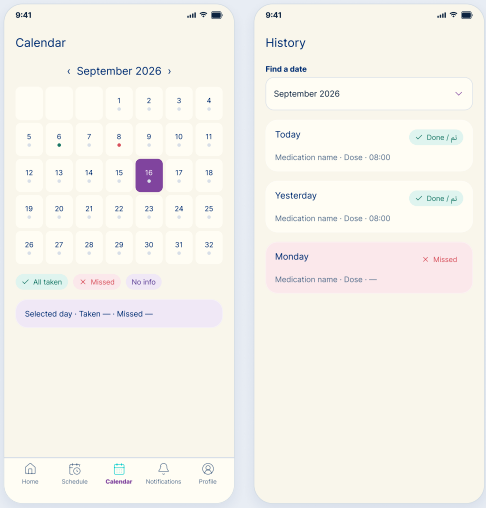
15 Reminder States

Full dose lifecycle with icon and label.



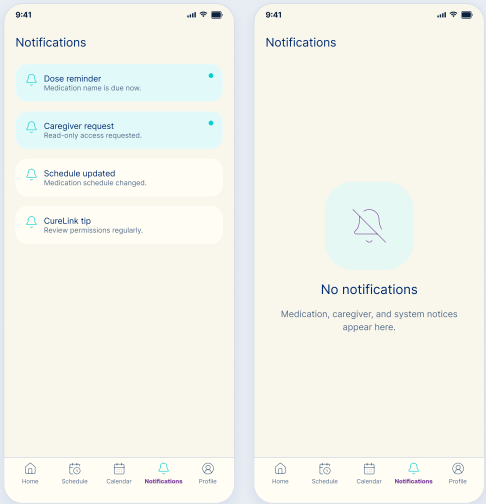
16 Calendar & History

Monthly state cells and generic daily logs.



17 Notifications

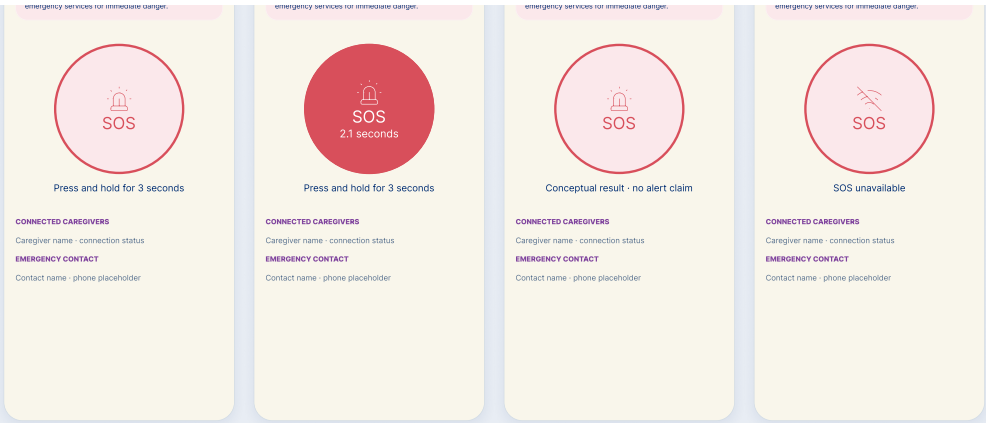
Categories, unread state, and empty state.



18 SOS

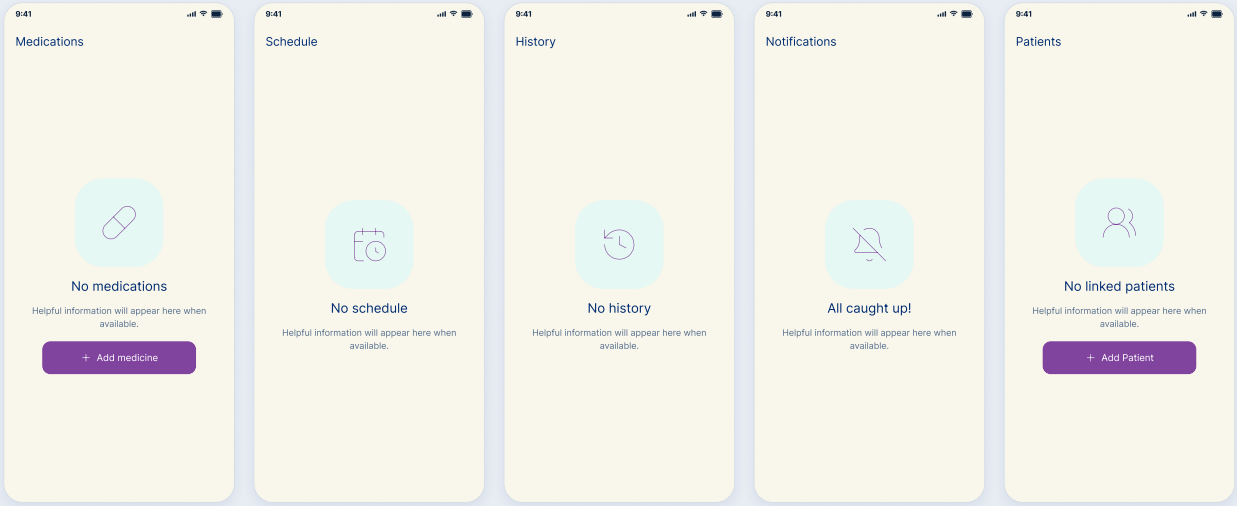
Distinct press-and-hold safety states.

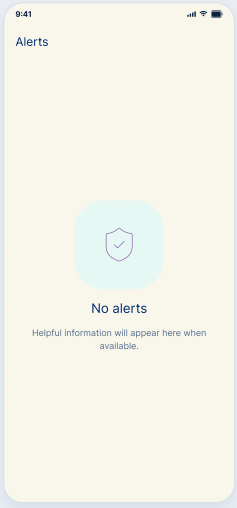




19 Empty States

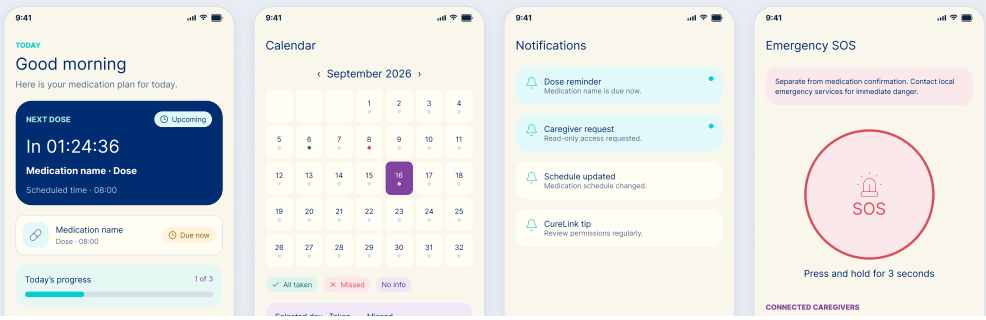
Persona-aware guidance and useful next steps.

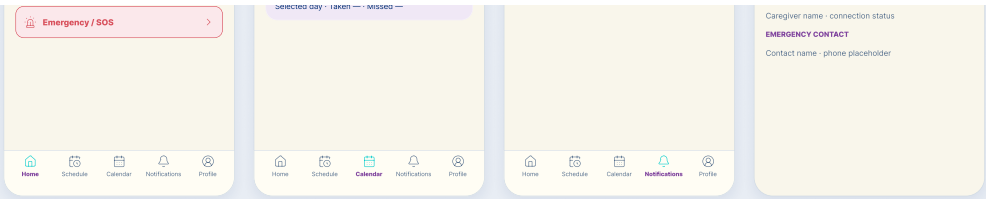




20 Light Mode

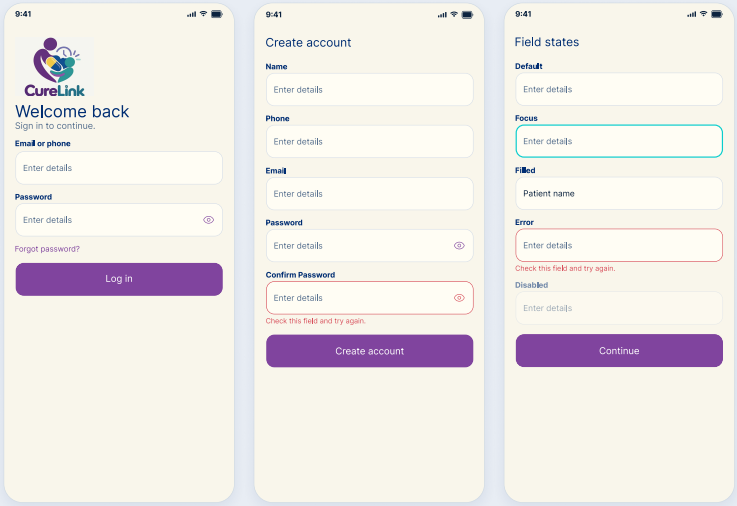
Warm Ivory with mint, purple, and aqua surfaces.





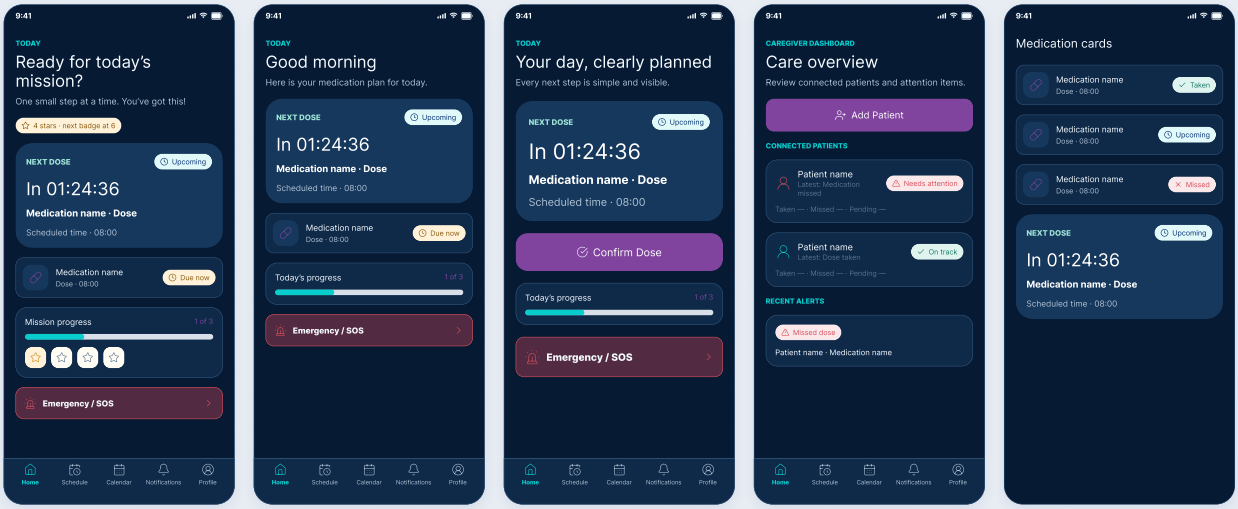
05 Authentication

Login, signup, and visual field states.



21 Dark Mode

Rich ink, blue-steel surfaces, and vibrant brand color.



Appendix B. Planning Assumptions & Validation Notes

Lecturer feedback

Only the team-reported positive initial reaction is included. Detailed verbal comments, formal grading, or lecturer-approved changes were not provided and are not invented.

Timeline

The official date ranges supplied by the team are reproduced. The Gantt graphic visualizes those workstreams at a high level; it is not a claim that every individual checklist item has a fixed date.

Trello ownership

The board export identifies the card owners listed in Section 1.7. The Week 2 ?Amira galal? card has no member ID in the export even though the title and checklist identify the work; the board should be updated to attach the owner explicitly.

Figma reference

The complete UI library supplied by the team is included in Appendix A. It is split across pages only to make the long Figma canvas readable on A4 pages.